**QP Code:25901386**

Mahatma Gandhi University, Kottayam

MGU-UGP(Honours) Examination March 2026

Core Course-MG4DSCMAT202- **Applied Mathematical Methods**

# Scheme of Valuation

**Part A**

**Answer any 5 questions**

**Each question carries 2 marks**

1. For correct roots — 2 Marks
   (either by factoring OR by any one of the prescribed methods)
2. f(x)=0 — 2 Marks
   (Give 1 mark for writing the approximation formula)
3. For correct example of non-periodic function — 2 Marks
4. For correct definition — 1 Mark
   For showing $f(-x) = -x + 1 \neq -f(x)$. Not an odd function — 1 Mark
5. For writing $\mathcal{L}^{-1}\left(\frac{1}{s+a}\right) = e^{-at}$ — 2 Marks
6. For writing answer $\frac{1}{s(s+2)}$ — 2 Marks
7. Integrating $\frac{dx}{1} = \frac{dy}{-1}$ and getting $x + y = C_1$ — 1 Mark
   Integrating $\frac{dx}{1} = \frac{dz}{1}$ and getting $x - z = C_2$ — 1 Mark
8. The general solution of Lagrange's equation $Pp + Qq + R$ is $F(u, v) = 0$, where $u(x, y, z) = C_1$ and $V(x, y, z) = C_2$ are two solutions of the auxiliary equation $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$. — 2 Marks

**Part B**

**Answer any 5 questions**

**Each question carries 6 marks**

9. For writing the given equation in the form $x = \phi(x)$ — 2 marks
   For the first iteration — 2 marks
   For the second iteration — 1 mark
   For correct answer — 1 mark

10. For the approximation formula — 2 marks
    For the first approximation — 1 mark
    For the second approximation — 1 mark

For the third approximation 1 mark

For correct answer 1 mark

11. For getting $a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} kdx = k.$ 3 Marks

For getting $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)cosnxdx = \frac{1}{\pi}\int_{-\pi}^{\pi} k.cosnxdx = 0$ 3 Marks

12. For getting $a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} 1dx = 1$ 2 Marks

For getting $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)cosnxdx = \frac{1}{\pi}\int_{-\pi}^{\pi} 1.cosnxdx = 0$ 2 Marks

For writing $b_n = 0.$ 2 Marks

13. For writing $\boldsymbol{\mathcal{L}}(\sin\omega t) = \frac{\omega}{s^2+\omega^2}$ 2 Marks

$\boldsymbol{\mathcal{L}}(t.\sin\omega t) = -\frac{d}{ds}\left(\frac{\omega}{s^2+\omega^2}\right) = \frac{2\omega s}{(s^2+\omega^2)^2}$ 2 Marks

$\boldsymbol{\mathcal{L}}(f''(t)) = s^2\boldsymbol{\mathcal{L}}(f) - sf(0) - f'(0)$ 1 Mark

$f(0) = 0, f'(0) = 0$, Thus $\boldsymbol{\mathcal{L}}(f''(t))\frac{2\omega s^3}{(s^2+\omega^2)^2}$ 1 Mark

14. For writing $\boldsymbol{\mathcal{L}}(3.8te^{2.4t}) = 3.8\boldsymbol{\mathcal{L}}(\boldsymbol{e^{2.4t}t}) = 3.8\frac{1}{(s-2.4)^2}$ 6 Marks

15. $\frac{dx}{y} = \frac{dy}{-x} \rightarrow -xdx = ydy$. Integrating and getting $x^2 + y^2 = C_1$ 3 Marks

$\frac{dy}{-y} = \frac{dz}{z-2y}$ gives $zdy + ydz = 2ydy$ integration gives $yz - y^2 = C_2$ 3 Marks

16. For writing $\frac{dx}{y-1} = \frac{dy}{x-1} = \frac{dz}{z}$ 2 Marks

$(x-1)dx = (y-1)dy$ and integration gives $x^2 - y^2 - 2(x-y) = C_1$ 2 Marks

$\frac{dx-dy}{-(x-y)} = \frac{dz}{z}$ and integration gives $z(x-y) = C_2.$ 2 Marks

**Part C**

**Answer any 3 questions**

**Each question carries 10 marks**

17. For finding $f(a)$ and $f(b)$ 2 Marks

For writing the formula and performing first iteration 2 Marks

For second iteration 2 marks

For third iteration 1 mark

For fourth iteration 1 mark

For fifth iteration 1 mark

For correct answer 1 mark

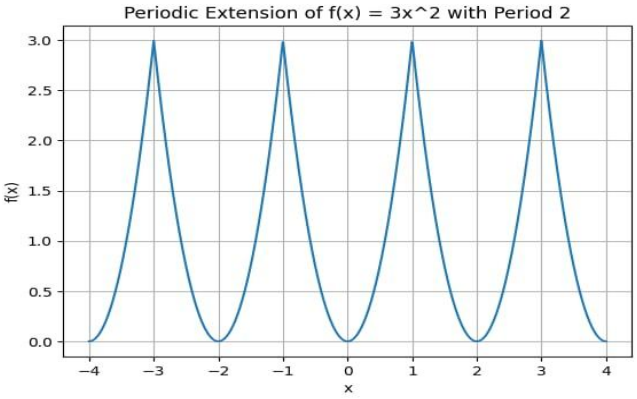
18. For writing $f(x) = a_0 + \sum_{n=1}^{\infty}(a_n cosn\pi x + b_n \sin n\pi x)$ 2 Marks

Getting $a_0 = \frac{1}{2\pi}\int_0^2 f(x)dx = 0$ 2 Marks

Getting $a_n = \frac{1}{\pi}\int_0^2 f(x)cosn\pi x dx = \frac{2}{n^2\pi^3}[(-1)^n - 1]$ 3 Marks

Getting $b_n = \frac{1}{\pi}\int_0^2 f(x)sinn\pi x dx = \frac{-2}{n\pi^2}$ 3 Marks

19.




2 Marks

$a_0 = \frac{1}{2}\int_{-1}^{1} 3x^2 dx = 1$ 3 Marks

$a_n = \int_{-1}^{1} 3x^2 cosn\pi x dx = \frac{12(-1)^n}{n^2\pi^2}$ 3 Marks

$b_n = 0$ 2 Marks.

20. For writing $\mathcal{L}(\cos\omega t) = \frac{s}{s^2+\omega^2}$ 5 Marks

For writing $\mathcal{L}(\sin\omega t) = \frac{\omega}{s^2+\omega^2}$ 5 Marks.

21. For writing $\mathcal{L}(t) = \frac{1}{s^2}$ and $\mathcal{L}(e^{-t}t) = \frac{1}{(s+1)^2}$ 5 Marks

For writing $\mathcal{L}(t) = \frac{1}{s^2}$ and $\mathcal{L}(\mathbf{3}.e^t.t) = \frac{3}{(s-1)^2}$ 5 Marks

22. (a) For getting $\frac{\partial z}{\partial x} = p = f'\left(\frac{xy}{z}\right)[\frac{zy-xyp}{z^2}]$ 2 Marks

For getting $\frac{\partial z}{\partial y} = q = f'\left(\frac{xy}{z}\right)[\frac{zx-xyq}{z^2}]$ 2 Marks

For getting $\frac{p}{q} = \frac{zy-xyp}{zx-xyq}$ and on simplification $z(xp - yq) = 0$ 1 Marks

(b) Differentiating partially w.r.t. x and getting $x - a = -zp$ 2 Marks

Differentiating partially w.r.t. y and getting $y - b = -zq$ 2 Marks

Eliminating a and b and getting $z^2(p^2 + q^2 + 1) = 1$ 1 Mark

**NB: Full marks can be awarded for any other correct answers, and appropriate stage marks may be determined by the examiners.**